Scenario C memo: Nobody on the flight deck during recovery

From crew rule to constraint. Rule R3 (brief D5) becomes one hold condition, `hold` below, on the ledger's flight-deck line: while it is true, landing clearance is withheld within the 200 ms safety-command ceiling (brief A.8). Hard limits are checked before any mission weight, so no D3 priority file lifts it; D5 sets by crew level what a human confirms, what is automatic and what is impossible. Each hold names its line in one bridge sentence.

The rule and why it exists

The brief's own example rule guards the marked landing zone on the 12 x 8 m deck (brief A.3): downwash from the 140 kg drone G4 throws loose gear, and a bad landing hurts whoever stands there (brief B, G4; research E3). We assume G4 is secured with no deck hand, so the fire party and landing officer (HLO) stand near but clear of the zone. People are the hazard and the only hands that clear it: 3-5 min at 14 aboard, 8-12 at 6, at 0 only if already clear (brief A.6). Below 20 % battery G4 must land within 12 min or ditch (brief B, G4). At 14 the rule is about people, at 6 about time, at 0 about evidence. The deck camera watches for loose or out-of-place gear; people on deck come from the HLO's report, and camera person counting is optional (WORKLOG DEC-23).

Trigger

R3 is on from the moment clearing must start for the zone to be empty before G4 has to land (from launch when the crew are too few to clear in time) until G4 is secured.

```
deadline = time G4 passed 20 % + 12 min        # brief B, G4
lead     = confirm_max + clear_max             # A.6: 1+5 min at 14, 3+12 at 6
recall   = first return order: D3 decision, sortie end, operator, silence  # brief D3; B, G4
opens    = launch if lead > 12 - approach or nobody aboard    # 6 and 0
           else min(recall, deadline - approach - lead)        # 14
window   = now >= opens and not G4.secured
people   = HLO deck report at 14 and 6; none at 0
deck_ok  = camera: markers in place, zone unchanged since the last sweep
           else HLO visual check (14, 6) else false
hold     = window and not (people == 0 and deck_ok)
```

Unknown never means clear. The HLO's visual check can replace a blind camera, never a fresh change detection; only a logged HLO override sets that aside (spray on the lens). Stale G4 telemetry: assume the earliest 20 % crossing the last reading allows. Unreported crew level: assume the next lower. Unanswered confirmation: no. A standing plan lapses when guests, priorities or crew level change, unless one for the new level was approved first; the departing crew approve the 0-aboard plan at handover.

Effect

- **Holds** landing clearance while `hold` is true, within 200 ms (A.8); no priority file lifts it.
- **Forbids** new tasks in the zone inside the window, except clearing and the sweep; the cards' own exclusions also apply: no winch (brief B, G2), so a 5 min cast must end before the window opens, and no ramp work (A.3).
- **Allows** landing once the zone is clear and the A.3 flight-deck limits hold.

Bridge sentence: "Recovery held: deck marker FD-2 is covered, so the landing zone is not clear (frame 0.4 s old); G4 at 18 % must land in about 10 min, needs 4 to approach, and confirm plus clear takes up to 6."

By crew level

In Scenario B, G4 reports 18 % at min 8. At our assumed drain of 100 % over 90 min it passed 20 % 1.8 min earlier, so it must land by min 18.2. The D3 decision at min 8 recovers G4, which is the recall, so clearing starts then: confirm 1 + clear 4 + approach 4 lands G4 at min 17 (demo seed, WORKLOG DEC-40). Without that order the window would open at min 8.2 at the latest: 10.2 min left, less our assumed

4 min approach, is 6.2 min against a 6 min lead at 14, a 12 s margin. At 6 the lead is 15 min against at most 8, so the zone stays clear from launch.

Aboard	A human confirms	Automatic	Impossible
14, patrol	Bridge confirms the clear-deck call in 15-60 s (A.6, applied by us); HLO may override. Hands: crew clear and sweep in 3-5 min	Clear-deck call when the window opens; hold and release	Landing on a detection nobody overrode
6, monitoring	Bridge approves a standing plan at launch, since a live yes plus clearing (3 + 12 min) cannot fit in 8; HLO may override	Zone clear from launch until G4 is secured; deck, winch and ramp work refused up to 90 min (endurance, brief B; our choice)	A recovery that needs clearing first
0, night transit	Nothing live (A.6); the departing crew's plan stands	Launch only if recovery stays possible for the whole 90 min sortie; land only on a fresh "unchanged, empty" reading	Clearing the deck; any confirmation or override

At 0 the hazard is loose gear, which the camera watches; a bad landing meets no fire team (research E3). At night the camera needs deck lighting or a low-light channel; thermal can add a hazard, never clear one. A blind camera means hold, so below 20 % G4 ditches, and the record states that cost. Our choice: the shore is told when a link exists, never asked, because the rule must work offline (SATCOM is zero in silence, A.5); research E3 and E8 assume shore clearance instead.

Live demo: 14, 6 and 0 aboard

The brief's should-have is the rule at two crew levels, shown live; we show three. Right after Scenario A, a short fixed segment replays Scenario B from minute 8 as the crew dial turns:

1. **14 aboard.** The bridge confirms the clear-deck call and the crew clear the zone. A marked item is then put over deck marker FD-2: recovery held within 200 ms with its bridge sentence; item removed: released.
2. **6 aboard.** No live call: the standing plan approved at launch is shown, and the log shows a deck work request at min 3, an event we added and label on screen, refused: "not now; when: after G4 is secured" (allowed at 14).
3. **0 aboard.** Webcam turned away: held, no confirm or override button, cost to G4 stated.

Assumptions

The rule's nine parameters (`rules.r3.*`: approach time, drain rate, staleness, change threshold, zone, override, camera, uncrewed launch, unchanged deck) live in `config/assumptions.yaml` with their reasons, and appear in the Physical Account.